

# THE MASTER DIAGRAM AND UNRAMIFIEDNESS OF $\mathcal{P}^{[d]_G}$ AT $\eta_C$

## 1. SETUP AND NOTATION

Let  $\mathcal{P} \rightarrow U$  be a  $G$ -torsor over  $U = C \setminus \mathfrak{m}$ , where  $G$  is a finite abelian group,  $H \trianglelefteq G$ , and  $C$  is a smooth projective curve. Fix  $p_\infty \in \mathfrak{m}$  and an integer  $d$  bounding the ramification of  $\mathcal{P} \rightarrow U$  at  $p_\infty$ .

Recall the four quotient schemes

$$\mathcal{P}^{[d]_H}, \quad \mathcal{P}^{[d]_G}, \quad (\mathcal{P}_{G/H})^{(d)}, \quad (\mathcal{P}_{G/H})^{[d]},$$

together with the canonical Cartesian square

$$\begin{array}{ccc} \mathcal{P}^{[d]_H} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{(d)} \\ \downarrow & & \downarrow q \\ \mathcal{P}^{[d]_G} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{[d]} \\ & \searrow G & \downarrow G/H \\ & & U^{(d)}, \end{array}$$

in which the arrows labelled  $H$ ,  $G/H$  and  $G$  are torsors under the indicated finite constant groups (in particular, *étale*), and the upper square is Cartesian (??).

The  $G/H$ -torsor  $\mathcal{P}_{G/H} \rightarrow U$  compactifies to a finite map of smooth projective curves  $f: C' \rightarrow C$ . Pick  $p'_\infty \in f^{-1}(p_\infty)$ , and let  $e := e(p'_\infty/p_\infty)$  be its ramification index over  $p_\infty$  (equivalently, the order of the inertia subgroup of  $p'_\infty$  in  $G/H$ ). Write

$$f^{(d)}: C'^{(d)} \rightarrow C^{(d)}$$

for the induced map of  $d$ -fold symmetric powers; it sends  $d \cdot p'_\infty \mapsto d \cdot p_\infty$ . Let

$$\pi_C: X_C \rightarrow C^{(d)}, \quad \pi_{C'}: X_{C'} \rightarrow C'^{(d)}$$

be the blowups at  $d \cdot p_\infty$  and  $d \cdot p'_\infty$  respectively, with exceptional divisors  $E_C, E_{C'}$  and generic points  $\eta_C, \eta_{C'}$ . The DVRs of these generic points are

$$V_C := \mathcal{O}_{X_C, \eta_C} \subset K(C^{(d)}), \quad V_{C'} := \mathcal{O}_{X_{C'}, \eta_{C'}} \subset K(C'^{(d)}).$$

The local relation  $t = u \cdot s^e$  between uniformizers at  $p_\infty, p'_\infty$  propagates to symmetric powers, giving

$$v_{C'}(x) = e \cdot v_C(x) \quad \text{for all } x \in K(C^{(d)})^\times,$$

i.e.  $V_{C'}/V_C$  is an extension of DVRs of ramification index exactly  $e$ .

## 2. THE MASTER DIAGRAM

To assemble the local torsor structures, their compactifications, and the blowups at infinity, we align the open geometry on the left, the projective/compactified geometry in the middle, and the birational geometry on the right.

$$\begin{array}{ccccccc}
 \mathcal{P}^{[d]H} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{(d)} & \xleftarrow{\text{open}} & C'^{(d)} & \xleftarrow{\text{blowup}} & X_{C'} \\
 \downarrow & & \downarrow q & & \downarrow f^{(d)} & & \downarrow f_X \\
 \mathcal{P}^{[d]G} & \xrightarrow{H} & (\mathcal{P}_{G/H})^{[d]} & & & & \\
 & \searrow G & \downarrow G/H & & \downarrow & & \\
 & & U^{(d)} & \xleftarrow{\text{open}} & C^{(d)} & \xleftarrow{\text{blowup}} & X_C
 \end{array}$$

## 3. PROOF OF UNRAMIFICATION

**\*\*Goal:\*\*** Prove that  $\mathcal{P}^{[d]G}$  is unramified above  $\eta_C$ .

In the language of valuation rings, "unramified above  $\eta_C$ " means that the integral closure of the Discrete Valuation Ring (DVR)  $V_C$  in the function field  $K(\mathcal{P}^{[d]G})$  is an unramified extension of  $V_C$ .

**\*\*Step 1: Translate assumptions to local rings\*\*** Let  $V_C$  and  $V_{C'}$  be the DVRs corresponding to  $\eta_C$  and  $\eta_{C'}$ . The text establishes that  $V_C \hookrightarrow V_{C'}$  is an extension of DVRs with ramification index  $e$ .

\* We are given that  $(\mathcal{P}_{G/H})^{[d]}$  is unramified over  $\eta_C$ . Therefore, localizing at a prime  $w$  over  $\eta_C$ , the corresponding local ring  $A_L$  of  $(\mathcal{P}_{G/H})^{[d]}$  is an unramified DVR over  $V_C$ . \* Let  $B$  be the local ring of  $\mathcal{P}^{[d]G}$  at a prime extending  $w$ .  $B$  is a DVR that extends  $A_L$ . Let its ramification index over  $A_L$  be  $r$ .

**\*\*Step 2: Leverage the Cartesian square\*\*** We are given that the upper square is Cartesian in the category of schemes:

$$\mathcal{P}^{[d]H} \cong \mathcal{P}^{[d]G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}$$

By passing to the local rings at the generic points of the exceptional divisors, this Cartesian property implies that the local ring for  $\mathcal{P}^{[d]H}$  over  $V_{C'}$  is exactly the tensor product:

$$\mathcal{O}_{\mathcal{P}^{[d]H}, \text{local}} \cong B \otimes_{A_L} V_{C'}$$

**\*\*Step 3: Apply the normality condition\*\*** Because  $\mathcal{P}^{[d]H}$  is constructed as a quotient of a regular scheme by a finite group  $(K_H \rtimes S_d)$ , it is a normal scheme. Therefore, the local ring  $B \otimes_{A_L} V_{C'}$  must be a normal domain. For a 1-dimensional local ring, being normal is equivalent to being a regular local ring (i.e., a DVR).

**\*\*Step 4: Analyze the tensor product singularity\*\*** We have two DVRs,  $B$  and  $V_{C'}$ , extending a common base DVR  $A_L$ .

\*  $V_{C'}$  has ramification index  $e$  over  $V_C$  (and thus over  $A_L$ , since  $A_L/V_C$  is unramified). \*  $B$  has ramification index  $r$  over  $A_L$ .

If we let  $\pi_L$  be the uniformizer of  $A_L$ , we can write the completed tensor product locally as:

$$\hat{B} \hat{\otimes}_{A_L} \hat{V}_{C'} \cong k[[x, y]]/(x^r - \pi_L, y^e - \pi_L) \cong k[[x, y]]/(x^r - y^e)$$

For the ring  $k[[x, y]]/(x^r - y^e)$  to be a regular local ring (which is required for normality), the curve  $x^r = y^e$  must be non-singular at the origin. A singularity of the form  $x^r = y^e$  is non-singular if and only if  $r = 1$  or  $e = 1$ .

**\*\*Step 5: Conclude unramification\*\*** If  $e = 1$ , the extension is trivially unramified. Assuming the general case where  $e > 1$  (ramification occurs at infinity), the normality of the Cartesian product strictly forces  $r = 1$ .

Since  $r = 1$ , the local ring  $B$  is an unramified extension of  $A_L$ . Because  $A_L$  is unramified over  $V_C$ , the composition is unramified. Thus,  $\mathcal{P}^{[d]G}$  is unramified above  $\eta_C$ . ■

## REFERENCES

- [1] A. Grothendieck, *Revêtements Étales et Groupe Fondamental (SGA 1)*, LNM 224, Springer, 1971.
- [2] The Stacks Project Authors, *Stacks Project*, <https://stacks.math.columbia.edu>.